PYTHON PROGRAMMING | EXERCISE 05

Word Stemming with the Porter Stemmer

Reducing words to stem forms with NLTK

NLTK's PorterStemmer applies the classic Porter suffix-stripping algorithm. The resulting stem is useful for grouping related word forms, though it is not always a dictionary word.

Question

Use the Porter Stemmer algorithm to perform word stemming on a list of words using Python libraries.

Code

```
from nltk.stem import PorterStemmer

# Install NLTK once with: python -m pip install nltk
stemmer = PorterStemmer()
words = ["running", "runs", "easily", "fairness", "relational", "ponies"]

print("Word       -> Porter stem")
for word in words:
    print(f"{word:10} -> {stemmer.stem(word)}")
```

Output

```
Word       -> Porter stem
running    -> run
runs       -> run
easily     -> easili
fairness   -> fair
relational -> relat
ponies     -> poni
```

Output screenshot

